Adapter SDK and Insurer Integration Specification

1. Contract principles

Use the repository's frozen SDK and Zod schemas as source of truth. Do not create a competing contract. Each adapter must expose stable metadata, validate insurer-specific fields, execute only an authorized workflow, parse deterministically, return structured status/errors, and include sanitized fixtures and README.

2. Suggested normalized result

Align names with existing types:

- runId, quoteId, insurerId
- status
- premium in canonical integer paise or approved decimal-safe representation
- currency (INR for rupee quotes)
- source (`automated` or `manual`)
- retrievedAt
- product/plan label
- sum insured/coverage summary where returned
- warnings
- errorCode and safeMessage
- evidence reference (local reference only)

3. State model

Recommended states: pending, running, completed, login_required, otp_required, manual_action_required, manual_required, failed, cancelled, unsupported. If the existing frozen contract differs, use the frozen values and map UI wording accordingly.

4. Error taxonomy

`AUTH_REQUIRED`, `OTP_REQUIRED`, `MANUAL_ACTION_REQUIRED`, `PORTAL_UNAVAILABLE`, `NAVIGATION_TIMEOUT`, `FIELD_NOT_FOUND`, `VALIDATION_FAILED`, `PARSE_FAILED`, `QUOTE_NOT_AVAILABLE`, `RATE_LIMITED`, `UNSUPPORTED_FLOW`, `CANCELLED`, `STORAGE_ERROR`, `INTERNAL_ERROR`.

5. Adapter requirements

- Pure `parse.ts` (or equivalent) separated from browser driver.
- At least three sanitized fixtures where required by the task plan.
- Positive, missing-field and malformed/ambiguous cases.
- Vitest tests asserting exact normalized outputs.
- README: prerequisites, authorized workflow, required fields, known limitations, fixture provenance/date, manual fallback.
- Never log credentials, OTPs, session cookies or unredacted personal data.
- Never bypass CAPTCHA, OTP, MFA, rate limits, or portal access controls.

6. Insurer matrix — planning notes from supplied task list

Insurer	Portal notes	Release posture
HDFC ERGO	Flutter/canvas, coordinate input/OCR	Automated only if spike and permission pass; else manual
ICICI Lombard	Angular, datepicker, inline result	Candidate automated; validate
Niva Bupa	Angular/Ionic login	Candidate automated; validate
Tata AIG	eConnect V2, user type, username, OTP	Human-assisted flow; validate
Star Health	User ID/OTP; public quote path to investigate	Automated only if supported/permitted; otherwise manual
United India	jQuery/JSP, image CAPTCHA	Advisor-assisted only if permitted; otherwise manual
Bajaj Allianz	Keycloak SSO, image CAPTCHA	Advisor-assisted only if permitted; otherwise manual
New India Assurance	BaNCS, CAPTCHA/reCAPTCHA	Likely manual unless feasibility and permission are confirmed

7. Manual fallback

- Manual result records premium, plan, cover, date/time, source and optional evidence.
- Result and PDF show “Manual”.
- Manual value must not be represented as insurer-verified by the app.
- Validate amount and required fields; no default zero.
- Evidence files follow retention/deletion policy.

8. Portal change process

Reproduce using authorized test account/dummy data; capture sanitized evidence; update parser and fixtures; run unit/regression tests; verify permitted live behavior; document known limitations; ship via release process.